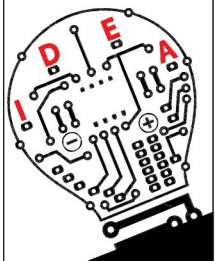
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Fig. 5: Light bulb with holder

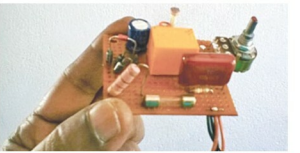


Fig. 7: Author's prototype

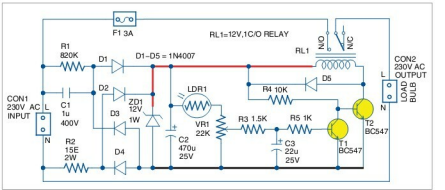


Fig. 6: Circuit of the smart bulb holder

12V DC low-current power supply and LDR-based twilight-controlled switching circuitry.

For the sake of simplicity, a capacitive potential divider circuit is used for the power supply. Rest of the circuit, as stated, is the ambient-light-controlled electromagnetic relay driver built around a standard photoresistor (LDR1) and two small general-purpose switching transistors T1 and T2.

In the circuit, 230V AC mains input is rectified by diodes D1 through D4 and limited by Zener diode ZD1 to 12V DC, then smoothed by capacitor C2. A 5mm photoresistor (LDR1) is used to detect the ambient light level.

When there is little or no light, the resistance of LDR1 is high. So the voltage on the base of T1 is very low, which drives the transistor into cut-off mode. Consequently, T2 conducts to energise relay RL1 and switch on the exterior light connected at CON2.

When sufficient light falls on the LDR1, the high voltage on base of T1 drives it into conduction. This interrupts the base current to T2, with the result that the relay de-energises. The switching level can be adjusted with the help of potentiometer VR1. Capacitor C3 provides a bit of hysteresis to prevent the system from jittering near the threshold level.

Since the circuit is connected directly to the AC mains power lines, house it inside an insulated cabinet made from wood/acrylic sheets. Make sure the relay is low-current type (well below 35 milliamperes). The author used a 12V, 400-ohm relay for the prototype (shown in Fig. 7).

This circuit is suitable for wall-mounting and/or pole installation. Connect 230V AC input to CON1 and take output from CON2 terminals.

Caution! This circuit works on 230V AC mains, so take care to avoid lethal shock. EFY



T.K. Hareendran is founder and promoter of TechNode Protolabz